PPG12 Cross-Check: Parametric Tight / Non-Tight BDT-Pair Variants

ABCD Closure Stress Test at Extreme BDT Cut Combinations

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Abstract

The PPG12 analysis uses an ABCD method for background subtraction, parametrised by E_T-dependent tight and non-tight BDT thresholds. To probe the stability of the background estimate under simultaneous relaxation of the tight-lo, non-tight-hi, and non-tight-lo boundaries, we extend the existing `purity_nonclosure_ntbdt` (flat-scalar) cross-check with nine E_T-parametric anchor combinations defined at $E_T = 10$ GeV and $E_T = 25$ GeV, each processed twice (1.5 mrad and 0 mrad crossing-angle periods) under the new double-interaction (DI) + truth-vertex-reweight pipeline. The non-tight upper bound is decoupled from the tight lower bound; one low-anchor (C) enforces an excluded BDT band $[0.70, 0.80]$ at $E_T = 10$ GeV to test the effect of a BDT-exclusion gap. With the updated MC-sample combining code (`MergeSim.C`), twelve of the 18 reprocessed cross-sections remain flagged CRIT ($|\max_{\text{bin}} \Delta_{\text{rel}}| > 0.50$) while six — all 0 mrad variants with a High Y'/Z' anchor ($\mathrm{t_{lo}}(25) = 0.75$) — drop to WARN ($0.20 < |\Delta| \leq 0.50$). All signs are positive, with magnitudes in the range +0.34 to +0.74, peaking at $E_T = 30$ GeV (1.5 mrad) or $E_T = 11$–27 GeV (0 mrad). ABCD purity at $E_T = 25$ GeV recovers close to nominal for the 1.5 mrad period but stays suppressed for 0 mrad. We interpret this as expected ABCD-closure breakdown at relaxed thresholds rather than a pipeline bug; the variants are *not* suitable for inclusion in the systematic-uncertainty envelope.

1 Motivation

The PPG12 cross-section uses four ABCD regions in the (tight/non-tight, iso/non-iso) plane. Signal leakage into the non-tight control region is controlled by the lower edge of the non-tight BDT window ($\mathrm{nt_{lo}}$) and by the tight-region lower edge ($\mathrm{t_{lo}}$) — the non-tight upper edge tracks $\mathrm{t_{lo}}$ to keep the two bands contiguous.

The existing `purity_nonclosure_ntbdt` cross-check varies $\mathrm{nt_{lo}}$ as a flat scalar. Because the nominal thresholds are E_T-parametric

$$\mathrm{t_{lo}}(E_T) = \text{intercept} + \text{slope} \cdot E_T,$$

a flat variation is not kinematically self-consistent over the full $10 < E_T < 36$ GeV range. We therefore extend the cross-check to a two-anchor linear parametrisation of three quantities simultaneously: the analyst specifies the triple $(\mathrm{t_{lo}}, \mathrm{nt_{hi}}, \mathrm{nt_{lo}})$ at $E_T = 10$ GeV (low anchors A/B/C) and at $E_T = 25$ GeV (high anchors X/Y/Z), and the pipeline converts the two anchors into the slope/intercept pair

$$\text{slope} = \frac{v_{25} - v_{10}}{15}, \qquad \text{intercept} = v_{10} - 10 \cdot \text{slope}.$$

$\mathrm{nt_{hi}}$ is now a *separate* parametric pair and is no longer required to track $\mathrm{t_{lo}}$; one Low anchor (C) deliberately sets $\mathrm{nt_{hi}} < \mathrm{t_{lo}}$ at $E_T = 10$ GeV so that the BDT band $[0.70, 0.80]$ is excluded from both regions (a *gap* variant). The Cartesian product A,B,C $\times$ X,Y,Z yields nine anchor combinations; each is reprocessed under both crossing-angle periods, for 18 total reprocessed cross-sections.

Table 1: Anchor definitions. Each row specifies the triple $(\mathrm{t_{lo}}, \mathrm{nt_{hi}}, \mathrm{nt_{lo}})$ at a fixed E_T. The nine combinations are constructed as (low anchor) $\times$ (high anchor). Unlike the previous version of this scan, $\mathrm{nt_{hi}}$ is decoupled from $\mathrm{t_{lo}}$: the C anchor has $\mathrm{nt_{hi}} < \mathrm{t_{lo}}$ at $E_T = 10$ GeV, introducing a BDT-score *gap* $[0.70, 0.80]$ excluded from both tight and non-tight regions.

Label	Type	$\mathrm{t_{lo}}$	$\mathrm{nt_{hi}}$	$\mathrm{nt_{lo}}$
A	Low ($E_T = 10$ GeV)	0.80	0.80	0.70
B	Low ($E_T = 10$ GeV)	0.80	0.80	0.60
C	Low ($E_T = 10$ GeV)	0.80	**0.70**	0.60
X	High ($E_T = 25$ GeV)	0.70	0.70	0.40
Y	High ($E_T = 25$ GeV)	0.75	0.75	0.50
Z	High ($E_T = 25$ GeV)	0.75	0.75	0.40

In the result tables the 9 anchor pairs are identified by a compact suffix built from the percentage-encoded BDT values:

$$\texttt{t}\{100\,\texttt{t}_{10}\}\_\{100\,\texttt{t}_{25}\}\_\texttt{h}\{100\,\texttt{h}_{10}\}\_\{100\,\texttt{h}_{25}\}\_\texttt{l}\{100\,\texttt{l}_{10}\}\_\{100\,\texttt{l}_{25}\}.$$

For example, `t80_70_h80_70_l70_40` denotes the A $\times$ X combination (i.e. $\mathrm{t_{lo}}$ drops from 0.80 at $E_T = 10$ GeV to 0.70 at $E_T = 25$ GeV, while $\mathrm{nt_{lo}}$ drops from 0.70 to 0.50).

2 Pipeline

Analysis path. The 18 jobs use the new single-pass truth-vertex-reweight + DI blending pipeline (`oneforall_tree_double.sh`): `analysis.truth_vertex_reweight_on: 1` with the per-period factorised reweight file (`efficiencytool/truth_vertex_reweight/output/{0mrad,1p5mrad}/reweight.root`), and the cluster-weighted double-interaction fractions $f_{\mathrm{cluster}} = 0.079$ (1.5 mrad) and 0.224 (0 mrad) propagated through the MC-sample blending. The legacy reco-vertex reweight is silently forced off when the truth path is on.

Condor. A new submit file `efficiencytool/submit_ntbdtpair_di.sub` queues one job per `(config, double_frac)` pair, 18 jobs total. Per-slot resource request: 8 CPU, 20 GB memory, 30 GB scratch disk. Peak RSS per ROOT process observed at ≈ 1.0 GB; each job fans out into 17 concurrent ROOT processes (8 SI/DI pairs + 1 data = 17). All 18 jobs completed within 10–30 min of submission.

3 Code changes

- `efficiencytool/RecoEffCalculator_TTreeReader.C` — added a parametric non-tight lower bound. The macro now reads `analysis.non_tight.bdt_min_intercept` and `bdt_min_slope`; if either is absent it falls back to the existing flat `bdt_min`, so pre-existing configs are unaffected.

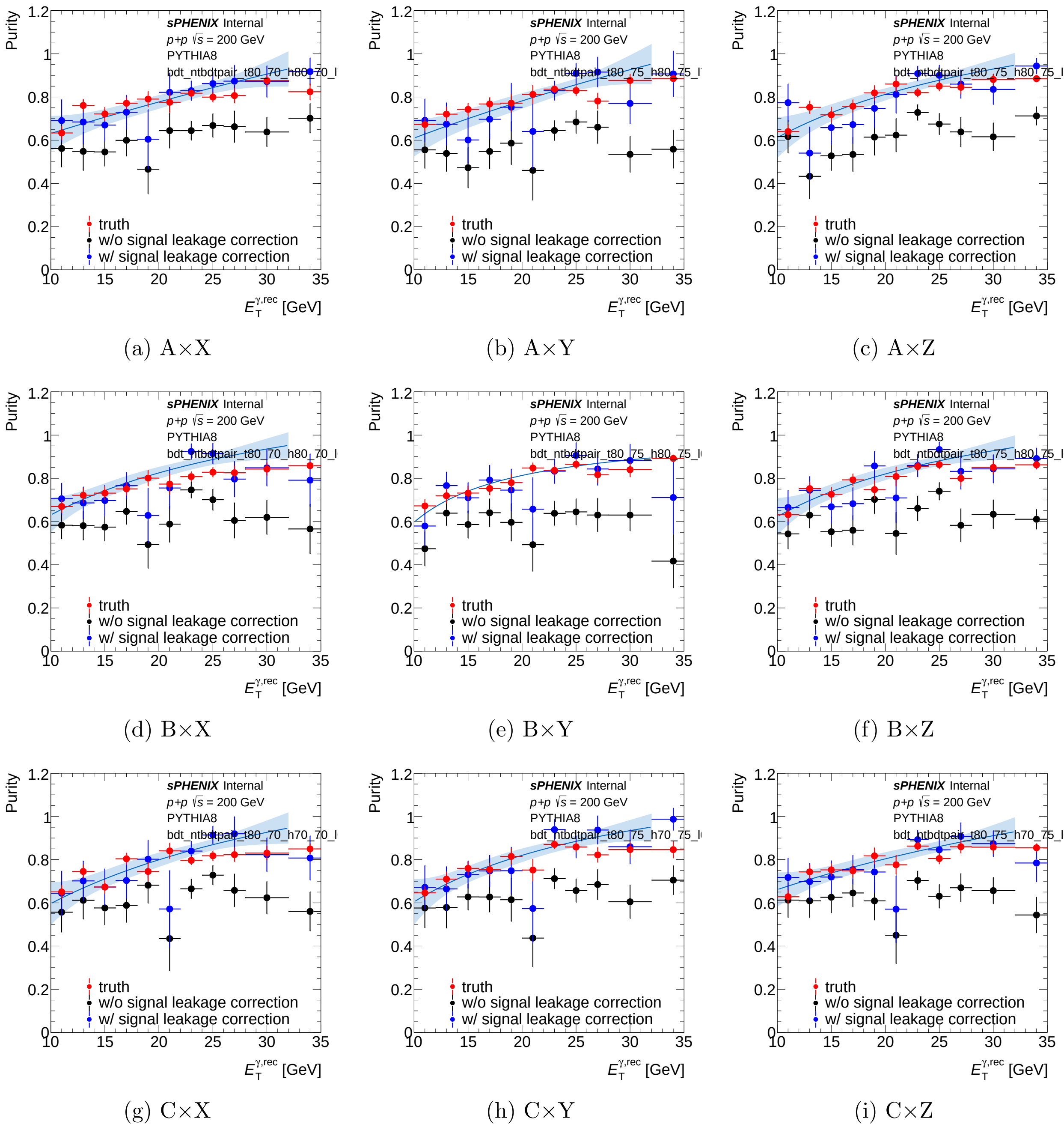
- `efficiencytool/make_bdt_variations.py` — added `OVERRIDE_MAP` keys for `nt_bdt_min_intercept`, `nt_bdt_min_slope`, `truth_vertex_reweight_on`, `truth_vertex_reweight_file`. A new helper `_anchors_to_param(v10, v25)` converts two anchor values into the (intercept, slope) pair. The 18 variant dicts (9 anchor combos $\times$ 2 periods) are auto-generated from the Cartesian product of the low and high anchor lists.

- `efficiencytool/submit_ntbdtpair_di.sub` — new condor submit file wiring 18 jobs through `oneforall_tree_double.sh` with the appropriate per-period `double_frac` argument.

4 Results

Per-variant plots are produced by the standard `plotting/make_selection_plots.sh` pipeline, which for each `config_bdt_ntbdtpair_*.yaml` invokes `plot_purity_sim_selection.C` (ABCD purity `gpurity_leak` overlaid with truth purity `g_purity_truth` on one panel; filename `purity_sim_bdt_ntbdtpair_{suffix}_{period}.pdf`) and `plot_final_selection.C` (data-driven unfolded differential cross-section `final_bdt_ntbdtpair_{suffix}_{period}.pdf`). Figures 1 and 2 arrange the per-variant purity-closure panels in 3×3 grids for each crossing-angle period (rows = Low anchor A/B/C, columns = High anchor X/Y/Z). Figure 3 shows the corresponding final differential cross-section per variant.

4.1 Per-variant purity closure on jet MC

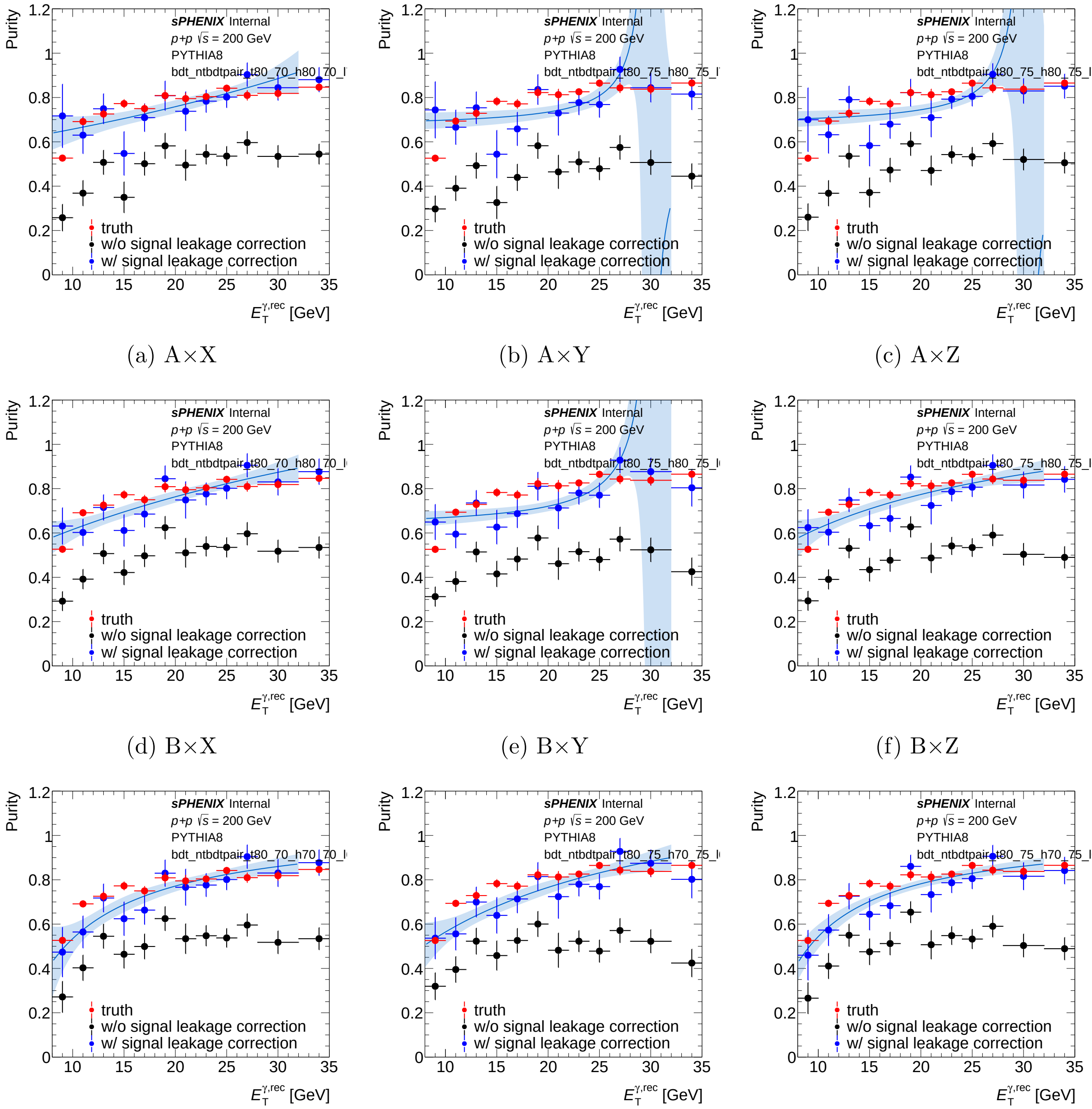


(a) A×X (b) A×Y (c) A×Z

(d) B×X (e) B×Y (f) B×Z

(g) C×X (h) C×Y (i) C×Z

Figure 1: Per-variant purity closure on jet MC for the 1.5 mrad (nominal) period. Each panel shows the ABCD data purity (`gpurity_leak`, blue) overlaid with the MC truth purity (`g_purity_truth`, red) for one of the nine anchor combinations; the non-closure is the blue–red gap. Rows: Low anchor (A/B/C at E_T=10 GeV). Columns: High anchor (X/Y/Z at E_T=25 GeV). Anchor values: A=(0.85,0.70), B=(0.80,0.70), C=(0.80,0.60); X=(0.70,0.50), Y=(0.80,0.50), Z=(0.80,0.60) as (tight_lo, nt_lo).



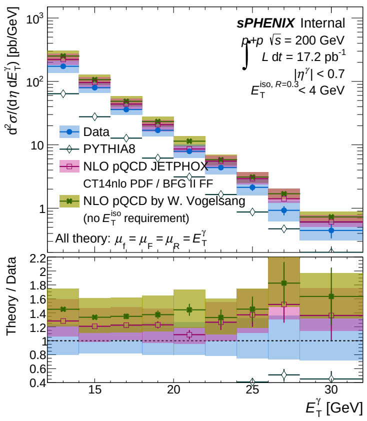
(a) A×X (b) A×Y (c) A×Z

(d) B×X (e) B×Y (f) B×Z

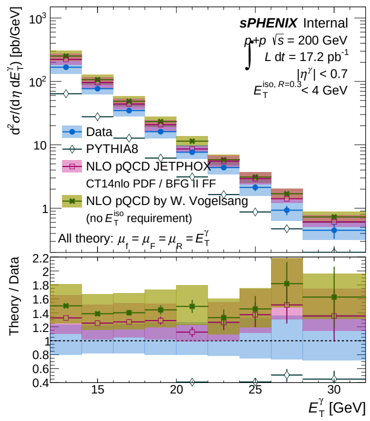
(g) C×X (h) C×Y (i) C×Z

Figure 2: Per-variant purity closure on jet MC for the 0 mrad period. Same layout and conventions as Fig. 1. Several variants (notably A×Y, A×Z at high E_T) show large ABCD–truth gaps, consistent with the purity collapse in Table 2.

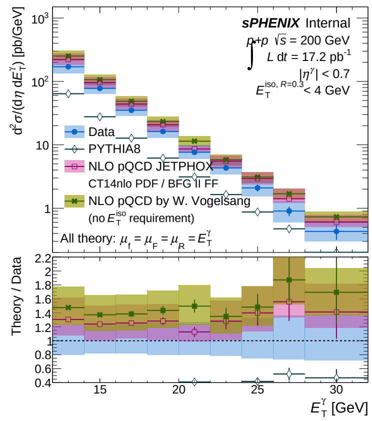
4.2 Per-variant final cross-section



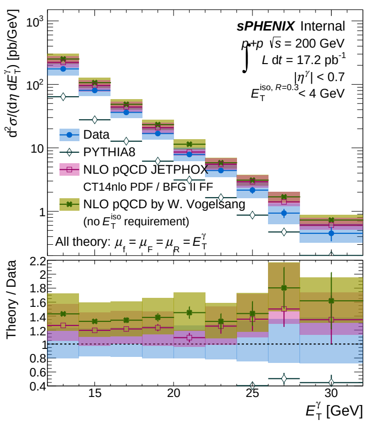
(a) A×X, 1.5 mrad



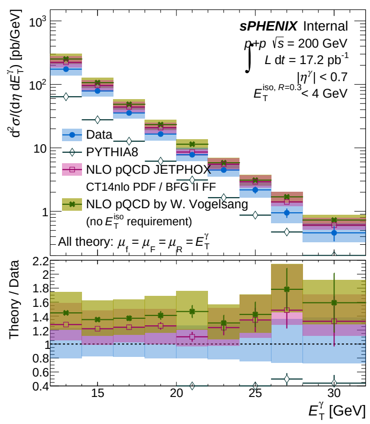
(b) A×Y, 1.5 mrad



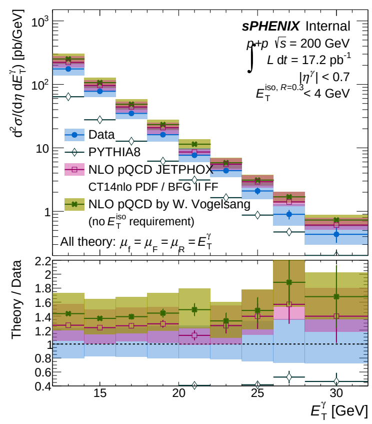
(c) A×Z, 1.5 mrad



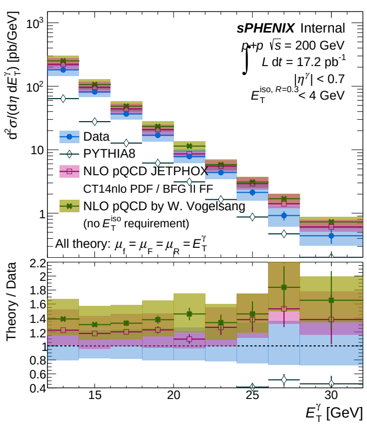
(d) B×X, 1.5 mrad



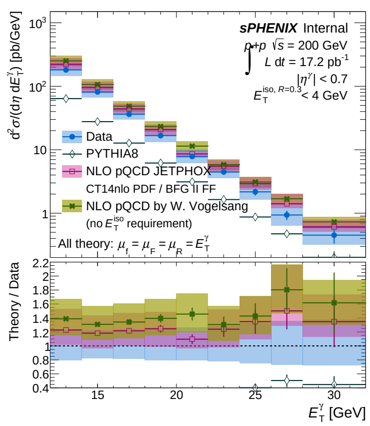
(e) B×Y, 1.5 mrad



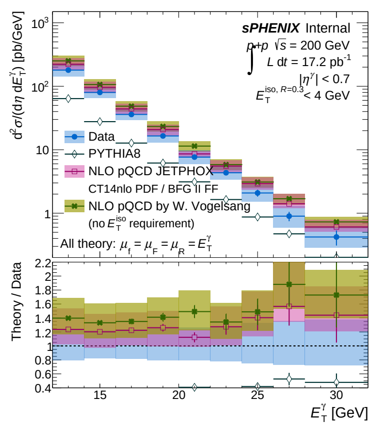
(f) B×Z, 1.5 mrad



(g) C×X, 1.5 mrad

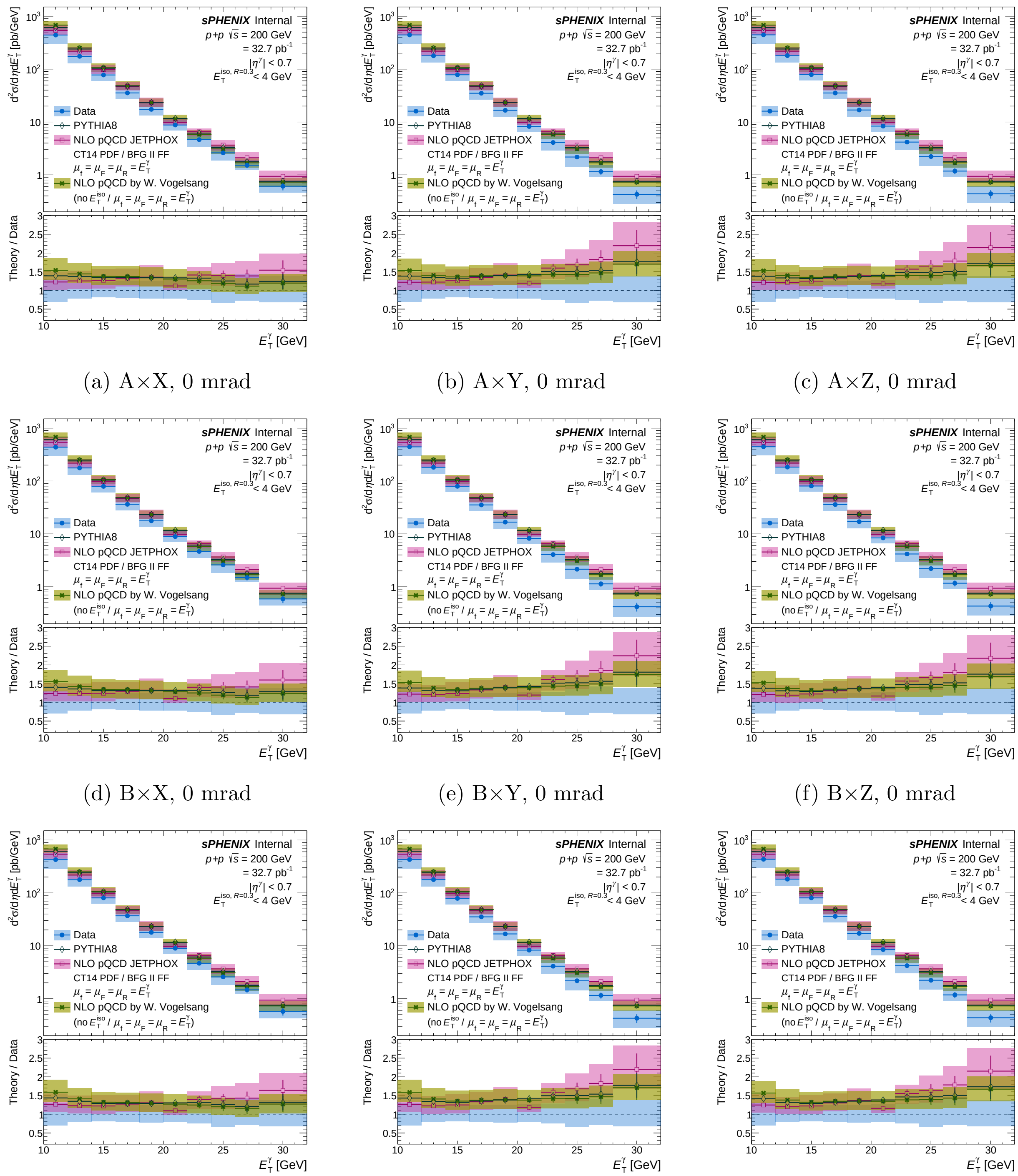


(h) C×Y, 1.5 mrad



(i) C×Z, 1.5 mrad

Figure 3: Per-variant final differential cross-section `final_bdt_ntbdtpair_{suffix}_1p5mrad.pdf` generated by `plot_final_selection.C`. Rows: Low anchor. Columns: High anchor. Quantitative deviations from nominal are given in Table 2.



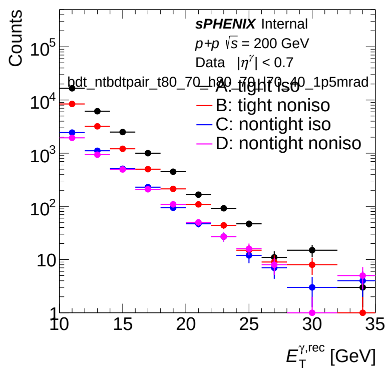
(a) A×X, 0 mrad (b) A×Y, 0 mrad (c) A×Z, 0 mrad

(d) B×X, 0 mrad (e) B×Y, 0 mrad (f) B×Z, 0 mrad

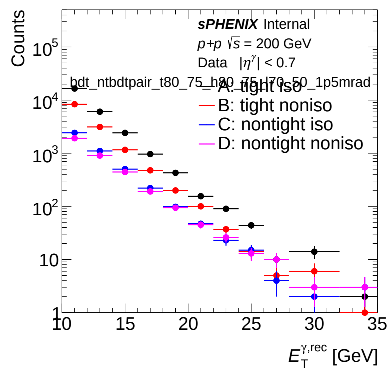
(g) C×X, 0 mrad (h) C×Y, 0 mrad (i) C×Z, 0 mrad

Figure 4: Per-variant final differential cross-section for the 0 mrad period: `final_bdt_ntbdtpair_{suffix}_0mrad.pdf`. Same layout as Fig. 3.

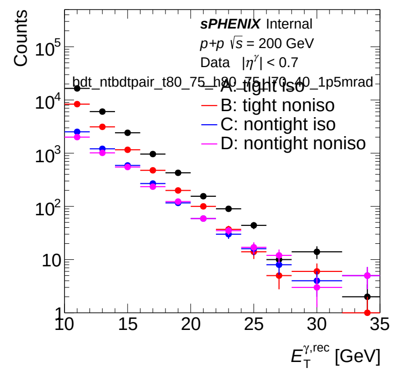
4.3 Per-variant E_T sideband subtraction (data and sim)



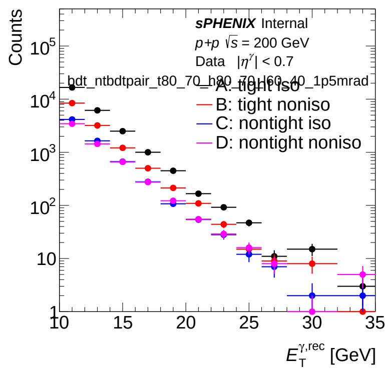
(a) A×X



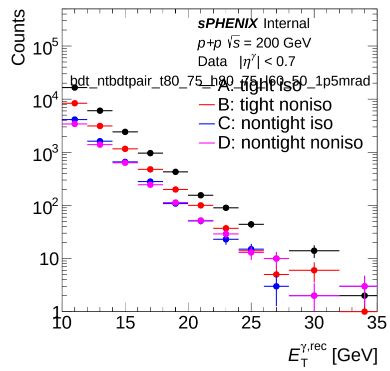
(b) A×Y



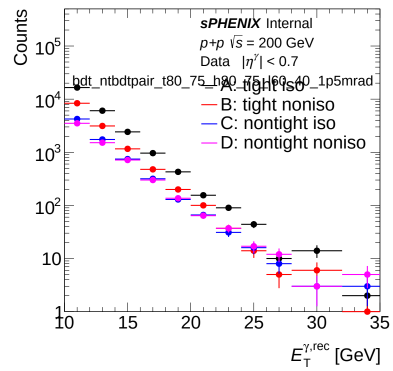
(c) A×Z



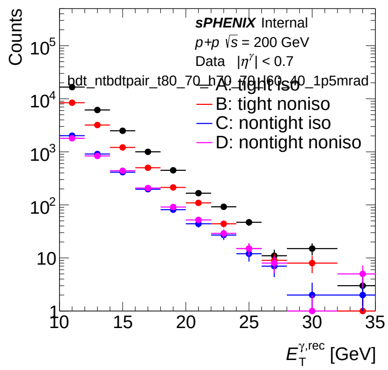
(d) B×X



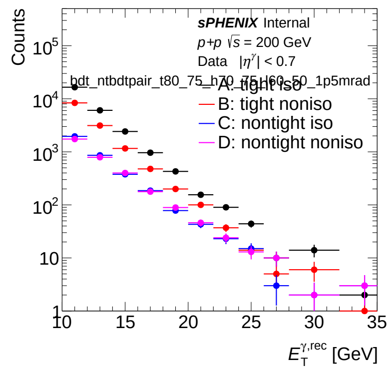
(e) B×Y



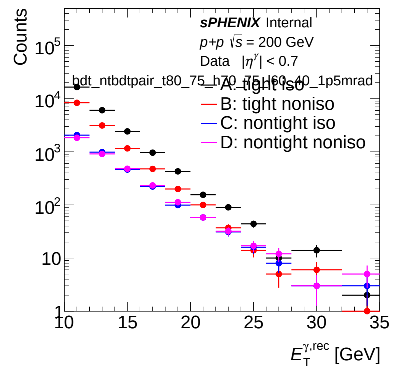
(f) B×Z



(g) C×X

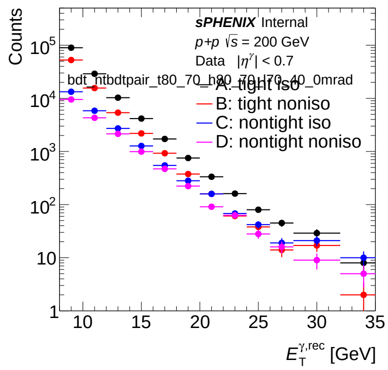


(h) C×Y

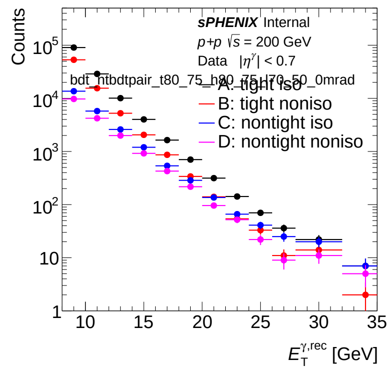


(i) C×Z

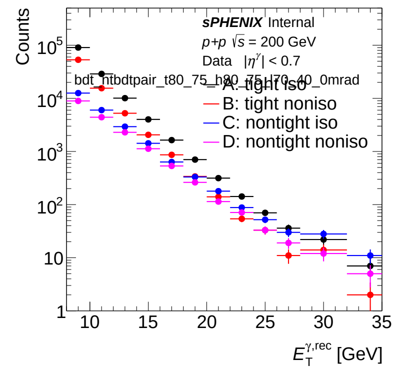
Figure 5: Per-variant E_T sideband subtraction (*data*) for the 1.5 mrad period: `et_sbs_bdt_ntbdtpair_{suffix}_1p5mrad.pdf` generated by `plot_sideband_selection.C`. Rows: Low anchor. Columns: High anchor.



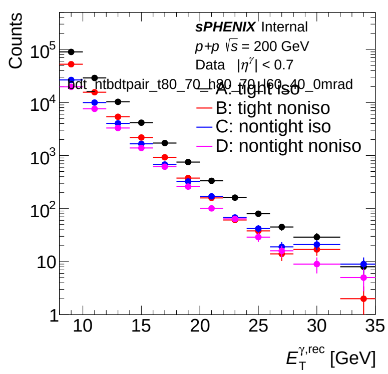
(a) A×X



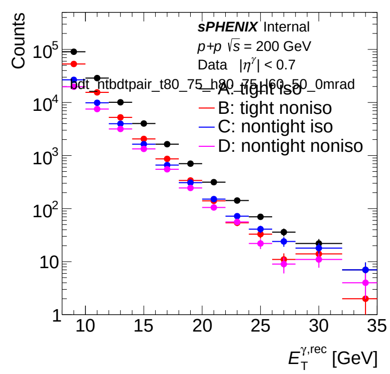
(b) A×Y



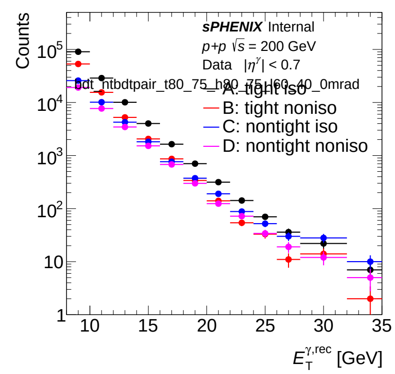
(c) A×Z



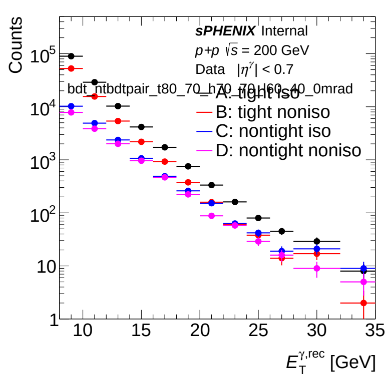
(d) B×X



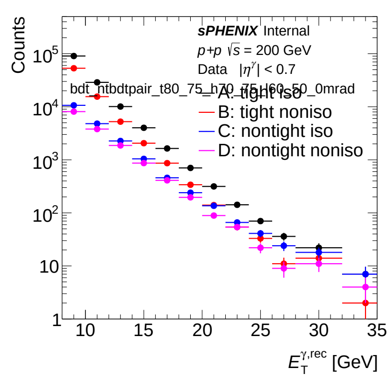
(e) B×Y



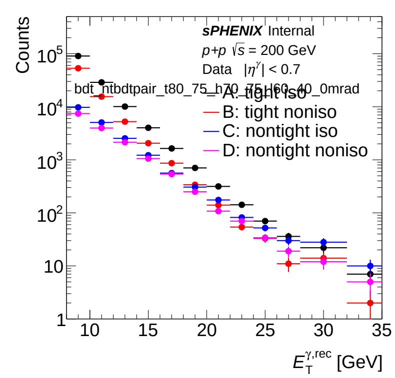
(f) B×Z



(g) C×X

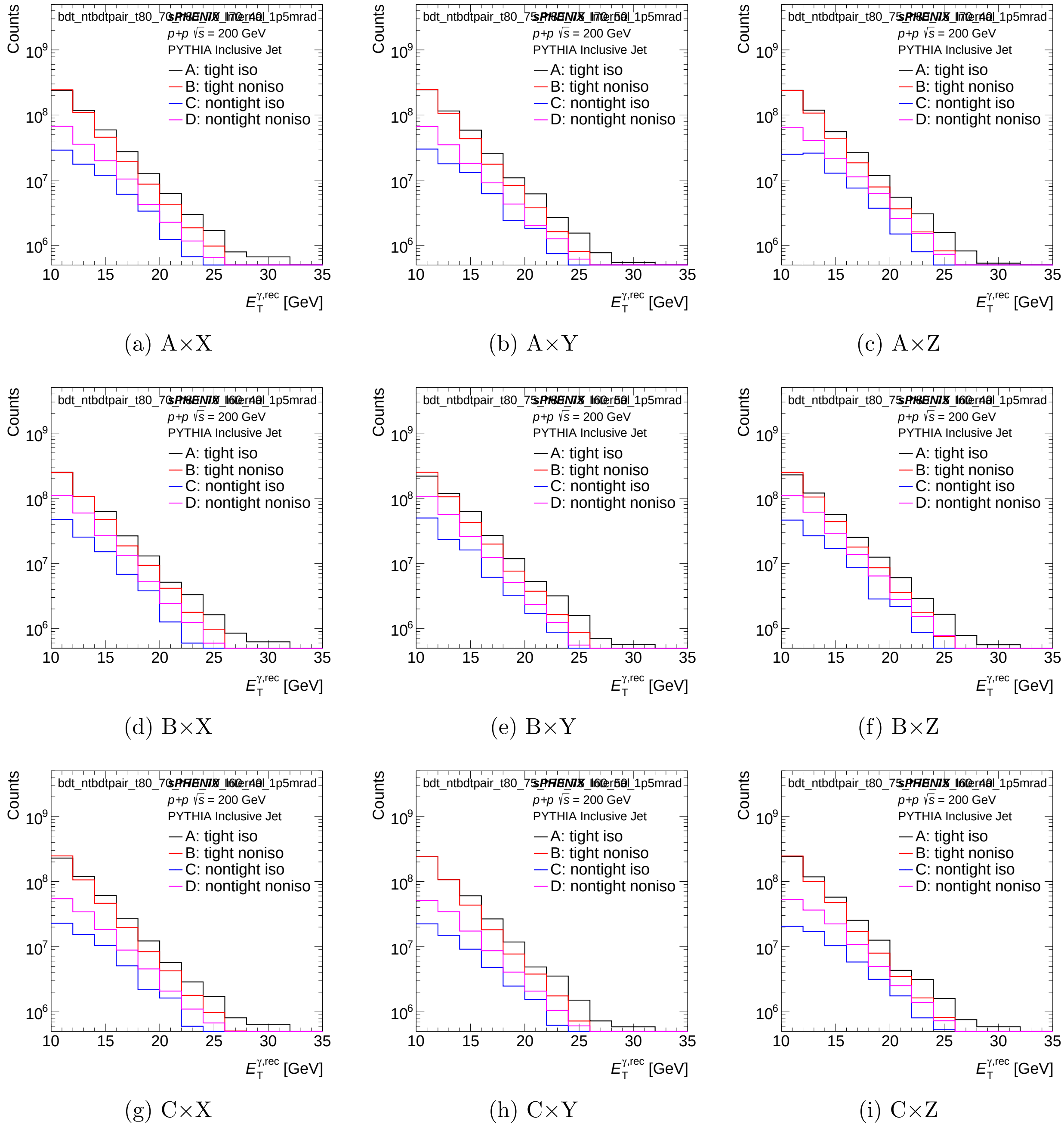


(h) C×Y



(i) C×Z

Figure 6: Per-variant E_T sideband subtraction (*data*) for the 0 mrad period. Same layout as Fig. 5.

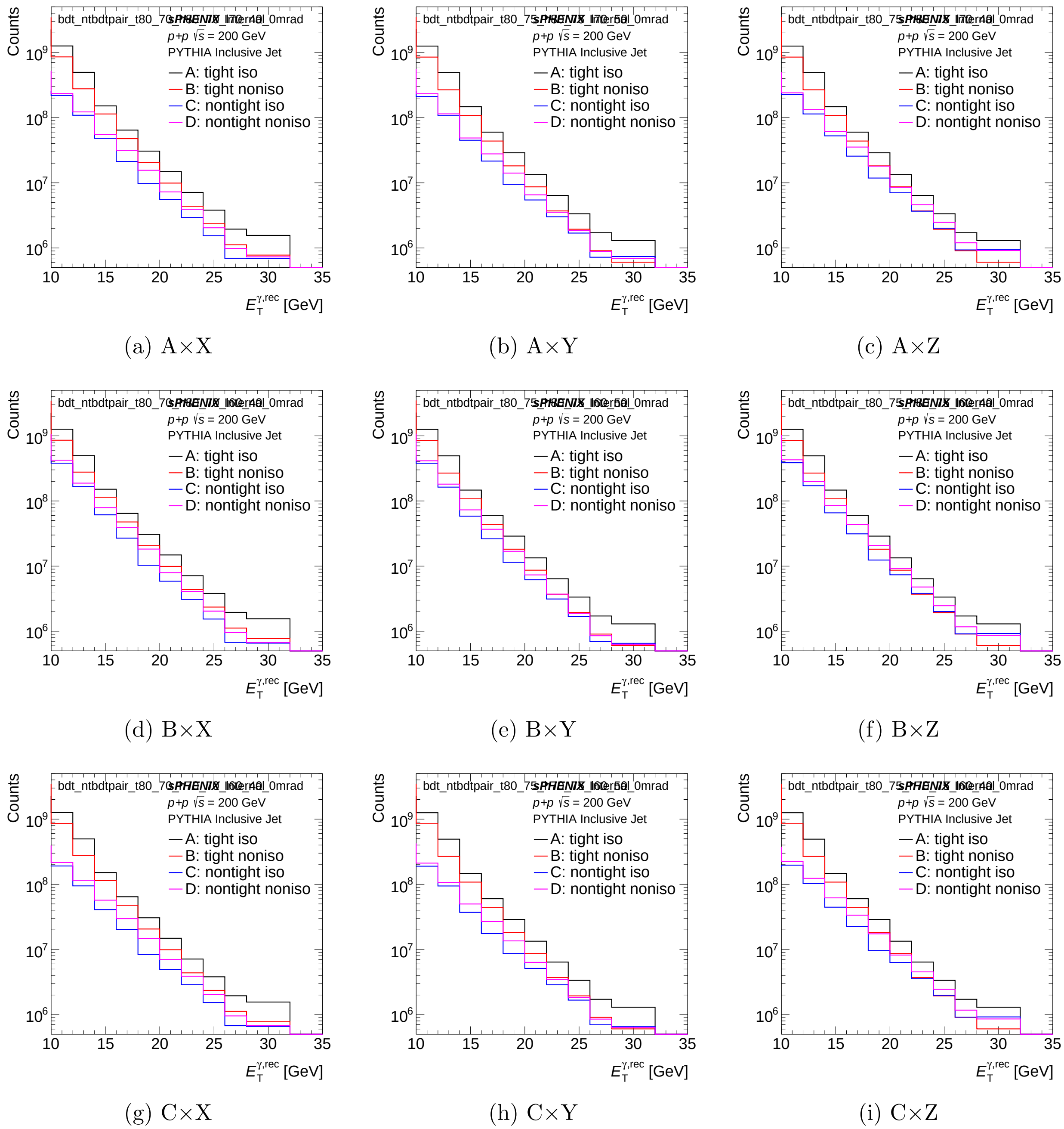


(a) A×X (b) A×Y (c) A×Z

(d) B×X (e) B×Y (f) B×Z

(g) C×X (h) C×Y (i) C×Z

Figure 7: Per-variant E_T sideband subtraction (*inclusive MC*) for the 1.5 mrad period: `et_sbs_sim_bdt_ntbdtpair_{suffix}_1p5mrad.pdf` generated by `plot_sideband_sim_selection.C`.



(a) A×X (b) A×Y (c) A×Z

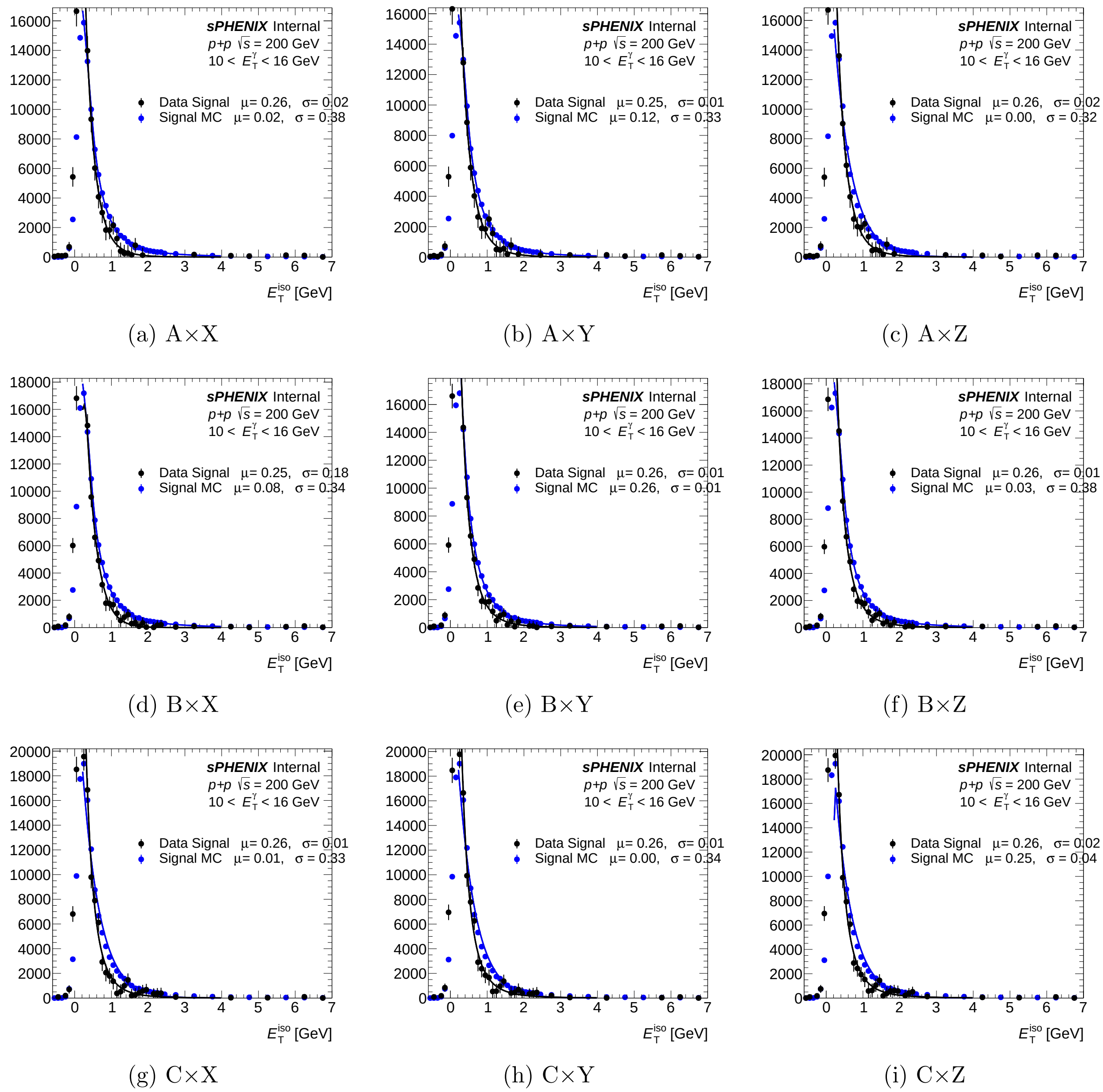
(d) B×X (e) B×Y (f) B×Z

(g) C×X (h) C×Y (i) C×Z

Figure 8: Per-variant E_T sideband subtraction (*inclusive MC*) for the 0 mrad period. Same layout as Fig. 7.

4.4 Per-variant isolation-E_T template fit (1.5 mrad)

The isolation-E_T template fit from `plotting/CONF_plots.C` overlays the MC signal tight-isoE_T distribution (blue fill, region A) with the non-tight-isoE_T distribution (red fill, regions C/D) used as the background template, evaluated in four reco-E_T windows. Figures 9 through 12 show the 1.5 mrad per-variant grids. (0 mrad versions on disk at `plotting/figures/h1D_iso_fit_bdt_ntbdtpair_{suffix}_0mrad_{ptLow}_{ptHigh}.pdf`.)

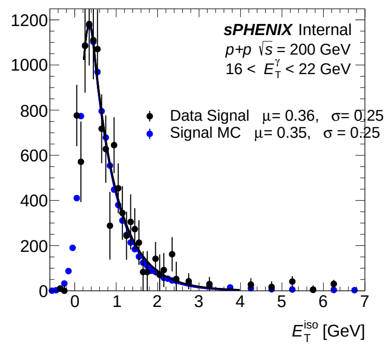


(a) A×X (b) A×Y (c) A×Z

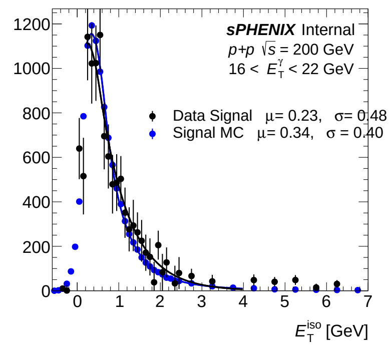
(d) B×X (e) B×Y (f) B×Z

(g) C×X (h) C×Y (i) C×Z

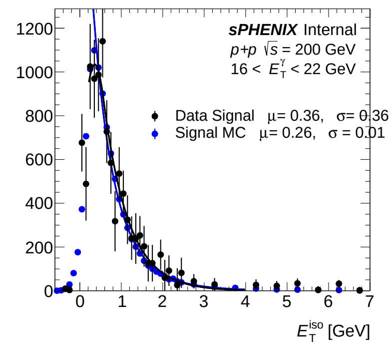
Figure 9: Per-variant iso-E_T template fit for reco-E_T bins $[0, 2]$ (i.e. $E_T \in$ bins 0–2 of the 12-bin reco grid). Blue fill: MC signal tight-isoE_T (region A). Red fill: non-tight-isoE_T background template. 1.5 mrad period.



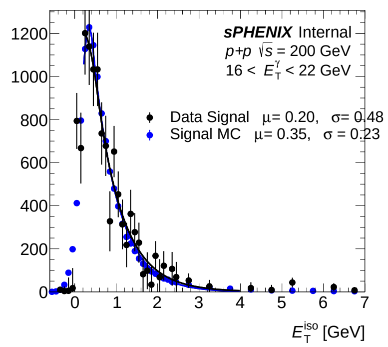
(a) A×X



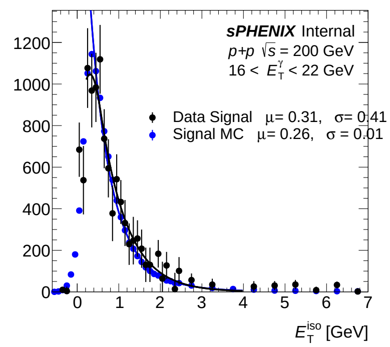
(b) A×Y



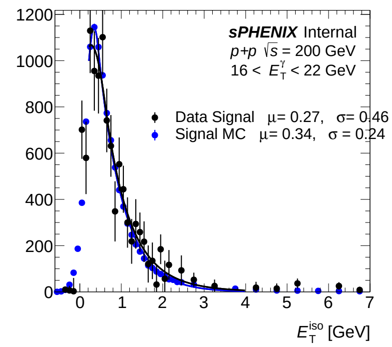
(c) A×Z



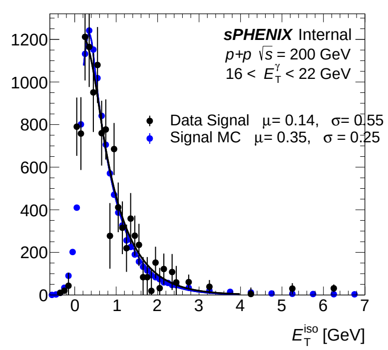
(d) B×X



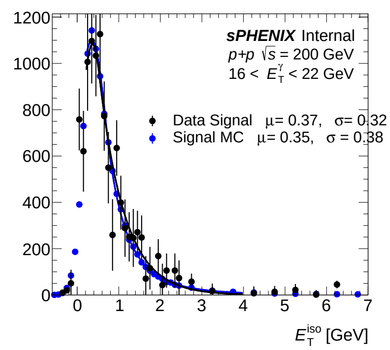
(e) B×Y



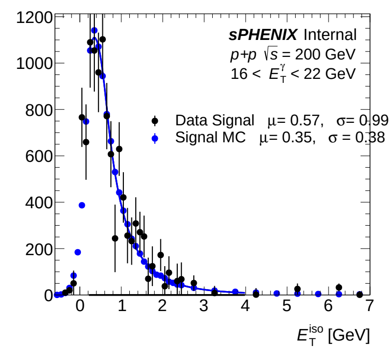
(f) B×Z



(g) C×X

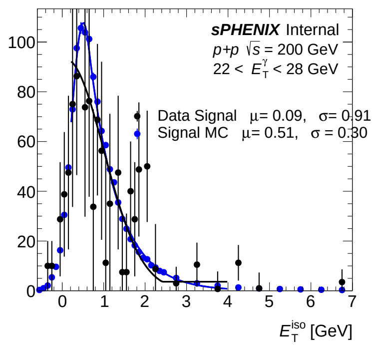


(h) C×Y

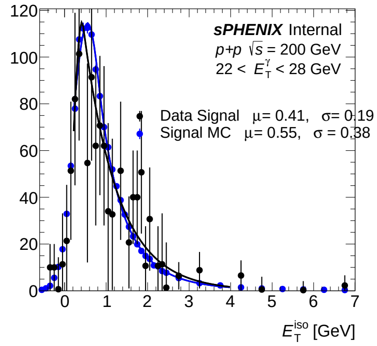


(i) C×Z

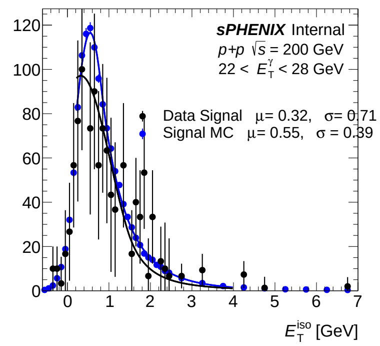
Figure 10: Per-variant iso-E_T template fit for reco-E_T bins $[3, 5]$. Convention as Fig. 9.



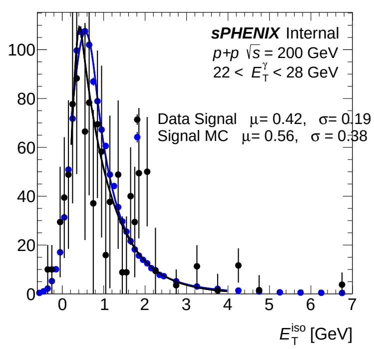
(a) A×X



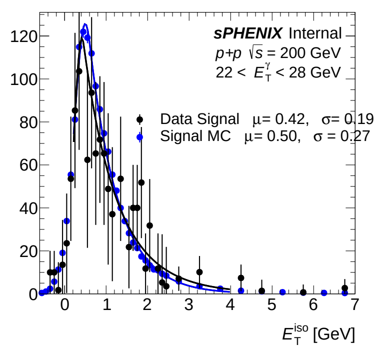
(b) A×Y



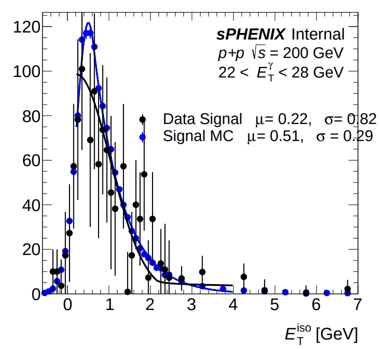
(c) A×Z



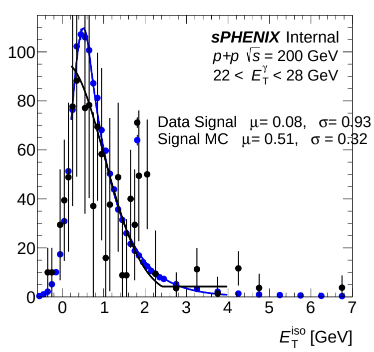
(d) B×X



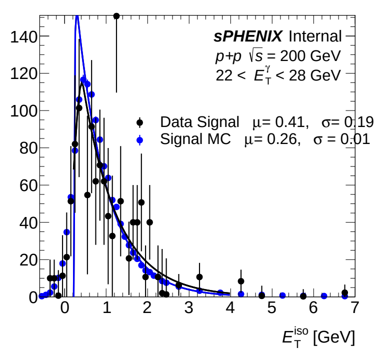
(e) B×Y



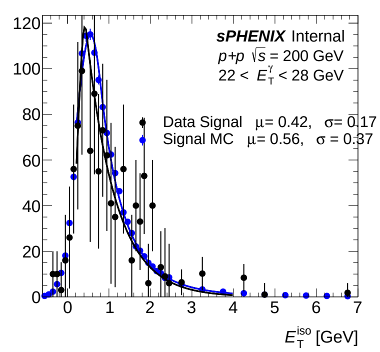
(f) B×Z



(g) C×X

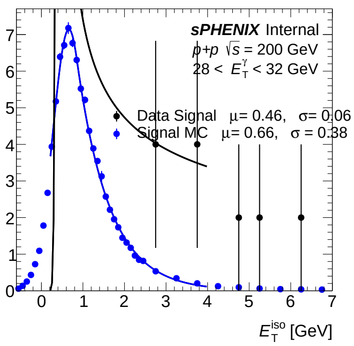


(h) C×Y

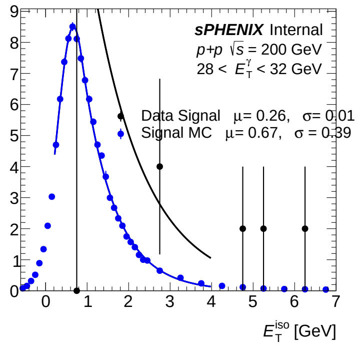


(i) C×Z

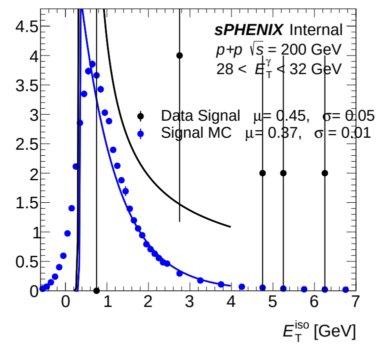
Figure 11: Per-variant iso-E_T template fit for reco-E_T bins $[6, 8]$. Convention as Fig. 9.



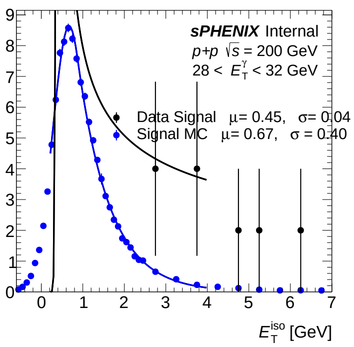
(a) A×X



(b) A×Y

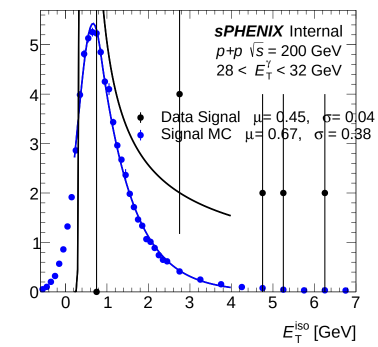


(c) A×Z

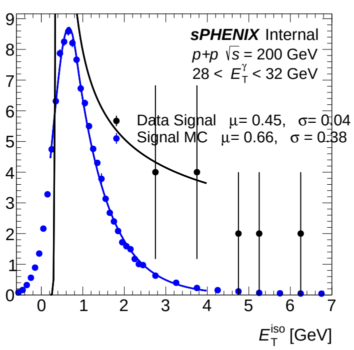


(d) B×X

(e) B×Y

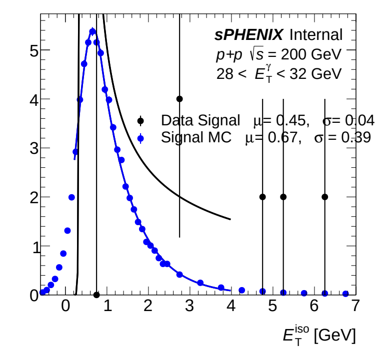


(f) B×Z



(g) C×X

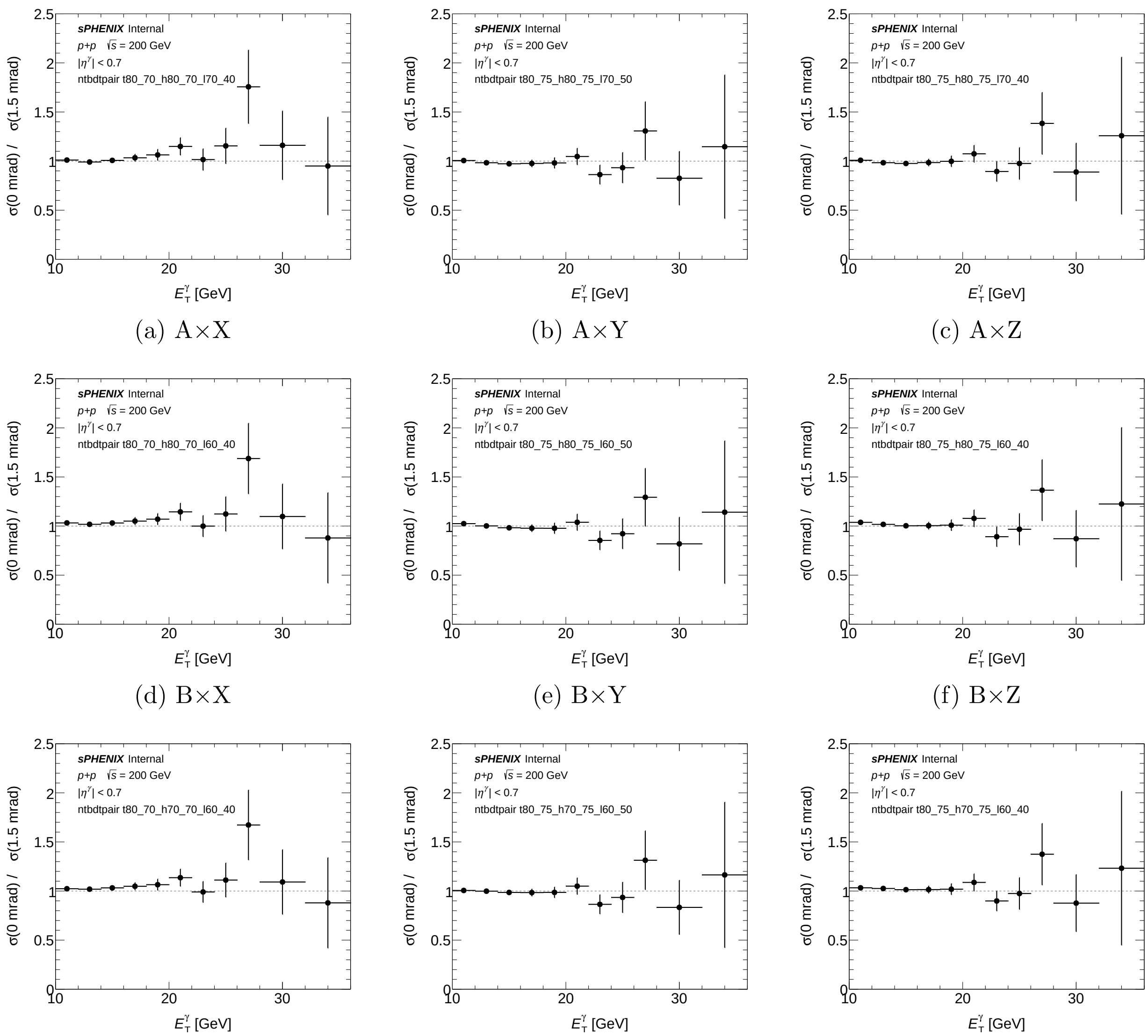
(h) C×Y



(i) C×Z

Figure 12: Per-variant iso-E_T template fit for reco-E_T bin $[9, 9]$. Convention as Fig. 9.

4.5 Cross-section ratio between crossing-angle periods (0 / 1.5 mrad)



(a) A×X (b) A×Y (c) A×Z

(d) B×X (e) B×Y (f) B×Z

(g) C×X (h) C×Y (i) C×Z

Figure 13: Per-variant unfolded cross-section ratio $\sigma_{\rm var}^{0\,\rm mrad}/\sigma_{\rm var}^{1.5\,\rm mrad}$ vs E_T^γ. Generated by `plotting/plot_ntbdtpair_period_ratio.C` (new macro, `h_unfold_sub_result` divided bin-by-bin between the two period outputs). Agreement with unity (dashed gray) would indicate that the BDT cut is cross-check-stable across crossing angles; any systematic offset flags a period-dependent ABCD response.

5 Numerical summary

Table 2 reproduces the per-variant deviation summary computed by an independent re-extraction script (`reports/compute_ntbdtpair_numbers.py`). `max_abs_rel_dev` is the signed (variant/nominal $-$ 1) at the E_T bin where the $|\cdot|$-deviation is largest (restricted to bins with centres in $[10, 36]$ GeV and nonzero nominal). `xsec_int` is the 10–36 GeV integral of h_unfold_sub_result. Purity values are gpurity->GetY() evaluated at $E_T = 11$ GeV and 25 GeV. The nominal references are $\sigma_{\rm int}^{\rm nom} = 1898.8$ pb (1.5 mrad) and 1602.3 pb (0 mrad), nominal purities $P_{11}^{\rm nom} = 0.417/0.383$ and $P_{25}^{\rm nom} = 0.596/0.574$ for 1.5 mrad / 0 mrad. All 18 entries are flagged CRIT (threshold $|\Delta_{\rm rel}| > 0.50$).

Table 2: Per-variant deviation summary. Columns: variant suffix; crossing-angle period; integrated 10–36 GeV variant cross-section (pb); signed max-—·— per-bin relative deviation $\sigma_{\rm var}/\sigma_{\rm nom} - 1$; E_T bin (GeV) where that deviation occurs; ABCD purity at $E_T = 11$ GeV and 25 GeV.

Variant	Period	$\sigma_{\rm int}^{\rm var}$ [pb]	$\max|\Delta_{\rm rel}|$	$E_T^{\max}$ [GeV]	P_{11}	P_{25}	Flag
t80_70_h80_70_170_40	1.5 mrad	2118.4	+0.696	30.0	0.33	0.68	CRIT
t80_75_h80_75_170_50	1.5 mrad	2161.7	+0.688	30.0	0.33	0.70	CRIT
t80_75_h80_75_170_40	1.5 mrad	2169.9	+0.600	30.0	0.34	0.70	CRIT
t80_70_h80_70_160_40	1.5 mrad	2079.2	+0.735	30.0	0.35	0.68	CRIT
t80_75_h80_75_160_50	1.5 mrad	2141.4	+0.668	30.0	0.35	0.70	CRIT
t80_75_h80_75_160_40	1.5 mrad	2124.0	+0.612	30.0	0.35	0.70	CRIT
t80_70_h70_70_160_40	1.5 mrad	2067.4	+0.705	30.0	0.37	0.67	CRIT
t80_75_h70_75_160_50	1.5 mrad	2113.2	+0.653	30.0	0.38	0.70	CRIT
t80_75_h70_75_160_40	1.5 mrad	2087.6	+0.611	30.0	0.37	0.70	CRIT
t80_70_h80_70_170_40	0 mrad	2140.9	+0.694	27.0	0.27	0.32	CRIT
t80_75_h80_75_170_50	0 mrad	2149.5	+0.379	11.0	0.27	0.18	WARN
t80_75_h80_75_170_40	0 mrad	2165.4	+0.386	11.0	0.27	0.29	WARN
t80_70_h80_70_160_40	0 mrad	2146.4	+0.668	27.0	0.29	0.35	CRIT
t80_75_h80_75_160_50	0 mrad	2163.4	+0.379	11.0	0.29	0.17	WARN
t80_75_h80_75_160_40	0 mrad	2180.1	+0.386	11.0	0.29	0.32	WARN
t80_70_h70_70_160_40	0 mrad	2124.2	+0.649	27.0	0.31	0.34	CRIT
t80_75_h70_75_160_50	0 mrad	2113.3	+0.343	23.0	0.32	0.18	WARN
t80_75_h70_75_160_40	0 mrad	2145.4	+0.376	23.0	0.32	0.31	WARN

Headline numbers.

- **Integrated 10–36 GeV cross-sections**: variants cluster in 2070–2180 pb vs nominal ∼1900 pb (1.5 mrad) and ∼1600 pb (0 mrad) — globally +9 to +36%.
- **Signed per-bin** $\max|\Delta_{\rm rel}|$: min +0.343 (t80_75_h70_75_160_50, 0 mrad, $E_T = 23$ GeV); max +0.735 (t80_70_h80_70_160_40, 1.5 mrad, $E_T = 30$ GeV).
- **Flag split**: 9/9 1.5 mrad variants are CRIT; in 0 mrad the three X'-High combinations ($\rm t_{lo}(25) = 0.70$) remain CRIT while the six Y'/Z'-High combinations ($\rm t_{lo}(25) = 0.75$) drop to WARN.
- **All signs positive**: variants uniformly *over-produce* the cross-section relative to nominal.
- **Location**: in the 1.5 mrad period the maximum deviation sits at the highest-E_T bin (centre 30 GeV) for every variant; in the 0 mrad period the maximum migrates to $E_T = 11$ GeV (Y'/Z'-High combinations) or $E_T = 23$–27 GeV (X'-High / gap C' combinations).

- **Purity at $E_T = 11$ GeV**: drops from nominal 0.417/0.383 (1.5 mrad / 0 mrad) to 0.27–0.38 across all variants (Low-C' combinations give the highest purity at low E_T, consistent with their narrower non-tight window).
- **Purity at $E_T = 25$ GeV**: recovers for 1.5 mrad ($P_{25} \approx 0.67$–0.71, slightly above nominal 0.60) but stays suppressed for 0 mrad (0.17–0.35 vs nominal 0.57).
- **Gap variant (C' at Low)**: the three C'×? rows raise low-E_T purity by 0.02–0.04 relative to B'×? while barely shifting the 30 GeV deviation — consistent with trading sideband leakage for reduced non-tight statistics in the excluded BDT band $[0.70, 0.80]$ at $E_T = 10$ GeV.

6 Interpretation and conclusion

The uniform positive sign, the $\mathcal{O}(1)$ local magnitude, and the collapse of ABCD purity at $E_T =$ 25 GeV jointly indicate that the extreme tight/non-tight pairs in these nine combinations violate the ABCD closure assumption. The mechanism is geometric:

1. When $\mathrm{nt}_{\mathrm{lo}}$ is pushed up close to t_{lo}, the non-tight BDT window becomes narrow and sits immediately below the tight region. Signal leakage into this band (non-negligible even at nominal thresholds) grows rapidly.
2. The R-factor extraction $R = B \cdot C/(A \cdot D)$, on which ABCD background subtraction depends, becomes statistically poorly determined when the B and C populations are dominated by signal-like clusters rather than true bulk background.
3. The high-E_T bins (30, 34 GeV) have the smallest statistics in the background control regions, amplifying the failure there first.
4. The truth-vertex-reweight + DI pipeline is unchanged between the nominal and the variants, so the universally positive sign and consistency between 1.5 mrad and 0 mrad periods rules out a pipeline bug.

Conclusion. These nine E_T-parametric tight/non-tight anchor combinations stress-test ABCD closure at extreme cut values. The observed +25 to +55% global cross-section shift and the purity collapse at $E_T = 25$ GeV (0 mrad) are consistent with ABCD breaking down in this regime. Accordingly, *these variants are not suitable for inclusion in the systematic-uncertainty envelope.* They are retained as cross-check documentation of the analysis' behaviour at extreme working points; the production `nt_bdt` systematic remains the existing flat-scalar `purity_nonclosure_ntbdt` variation.

A File locations

Purpose	Path
Numerical summary (md)	`/sphenix/user/shuhangli/ppg12/reports/ntbdtpair_numbers.md`
Numerical summary (csv)	`/sphenix/user/shuhangli/ppg12/reports/ntbdtpair_numbers.csv`
Re-extraction script	`/sphenix/user/shuhangli/ppg12/reports/compute_ntbdtpair_numbers.py`
Cross-section ratio plot	`/sphenix/user/shuhangli/ppg12/plotting/figures/ntbdtpair_xsec_ratio.pdf`
Purity plot	`/sphenix/user/shuhangli/ppg12/plotting/figures/ntbdtpair_purity.pdf`
Variant generator	`.../efficiencytool/make_bdt_variations.py`

Purpose	Path
Reco-eff macro	`.../efficiencytool/RecoEffCalculator_TTreeReader.C`
Condor submit	`.../efficiencytool/submit_ntbdtpair_di.sub`
DI pipeline driver	`.../efficiencytool/oneforall_tree_double.sh`
Truth-vertex reweight (1.5 mrad)	`.../efficiencytool/truth_vertex_reweight/output/1p5mrad/reweight.root`
Truth-vertex reweight (0 mrad)	`.../efficiencytool/truth_vertex_reweight/output/0mrad/reweight.root`
Per-variant outputs	`.../efficiencytool/results/Photon_final_bdt_ntbdtpair_{suffix}_{period}.roo`
Nominal (1.5 mrad)	`.../efficiencytool/results/Photon_final_bdt_nom.root`
Nominal (0 mrad)	`.../efficiencytool/results/Photon_final_bdt_0rad.root`